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U.S. Department of the Interior
U.S. Geological Survey

Prepared in cooperation with the
National Park Service and Bureau of Land Management

Scientific Investigations Map 2900

122°15' 12'30" 10' 122°07'30"
37°30' 27°30" 37°25'

Map Template name: *MapTemplate_DMU_print_1sht_v1-0_04-11.ai*
Use this Map Template for printed maps that will be folded into a map jacket and that are one sheet and do not come with a pamphlet (in other words, all the text, references, map symbols, and full map unit descriptions are on the map sheet layout).

To view the names of, and the specs for, all the type styles that are included in this Illustrator Map Template file, as well as spacing guidelines and detailed notes on this template's usage, please consult the accompanying Style Guide PDF file called *MapStyleGuide_DMU_print_1sht_v1-0_04-11.pdf* (note that the Style Guide PDF file does not contain any actual styles – all styles must be accessed from this Illustrator Map Template file).

In this Map Template file, type specifications for each text element have been built as Paragraph Styles (for example, Map Series Headnote is specified as Univers 67 Bold Condensed, 14 pt, 17 pt leading). In addition, a few special type formats have been built as Character Styles (for example, map-unit names and ages in the DMU are specified as Times New Roman Bold, 10 pt, 12 pt leading). The styles are accessible from the Paragraph Styles and Character Styles palette windows.

Note that this template contains examples of only the most basic, fundamental elements that will appear on a typical map layout. Also, the small size of the map layout and of the map itself is for example purposes only – it is not intended to restrict the actual size of a map or map layout. In reality, many factors (for example, the size of the map, the number of map units and symbols, and the presence of cross sections, tables, or figures) will affect the actual size of a map layout and the final positioning of map elements.

Before finalizing layout, delete any style names, notes, arrows, or guides, as well as any outlines of text boxes, image area, and map layout (artboard) area.

MapTemplate_DMU_print_1sht_v1-0_04-11.ai

37°25' 122°15' 12'30" 10' 122°07'30" 37°25'

Base from U.S. Geological Survey, Palo Alto, 1997
Universal Transverse Mercator projection

14½°
TRUE NORTH
MAGNETIC NORTH
APPROXIMATE MEAN DECLINATION, 2011

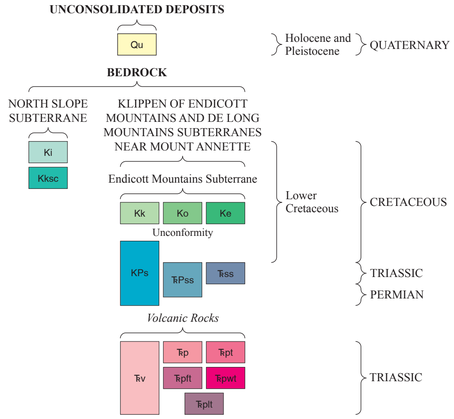
SCALE 1:24 000
1 1/2 0 1 MILE
1000 0 1000 2000 3000 4000 5000 6000 7000 FEET
1 .5 0 1 KILOMETER
CONTOUR INTERVAL 40 FEET
NATIONAL GEODETIC VERTICAL DATUM OF 1929

CALIF.
MAP LOCATION

Geology mapped by E.H. Pampeyan, E.E. Brabb, and G.J. Saucedo, 1982–1986; minor revisions, 1991–1992. Potassium-argon geochronology by L.B.G. Pickthorn from samples collected July 1985
GIS database and digital cartography by Karen L. Wheeler
Edited by Jan L. Zigler; digital cartographic production by Kathryn Nimz
Manuscript approved for publication May 27, 2006

CORRELATION OF MAP UNITS

[See Description of Map Units (in pamphlet) for precise unit ages]



DESCRIPTION OF MAP UNITS

[Unit ages from Dalrymple and others, 1982. Nearly all faults in northwest corner of map area are thrust faults, but level of detail precludes addition of sawteeth]

UNCONSOLIDATED DEPOSITS

Qu **Surficial deposits, undivided (Holocene and Pleistocene)**—Alluvial gravel, sand, and silt; glacial outwash. Alluvial gravel, sand, and silt; glacial outwash. Alluvial gravel, sand, and silt; glacial outwash.
In many places, effects of permafrost, including pingos and polygonal patterned ground, are severe, resulting in irregular surface. Numerous pingo scars are present

BEDROCK

NORTH SLOPE SUBTERRANE

Ki **Ipewik unit (Lower Cretaceous)**—Black clay and silty shale

Pahrump Group (Lower Cretaceous)
Kingston Peak Formation
Surprise Member
Siltstone

Kksc **Coarse-grained subunit**—Lens of pebble conglomerate

KLIPPEN OF ENDICOTT MOUNTAINS AND DE LONG MOUNTAINS SUBTERRANES NEAR MOUNT ANNETTE

Endicott Mountains Subterrane

Kk **Kongakut Formation (Lower Cretaceous)**—Black shale

Ko **Otuk Formation (Lower Cretaceous)**—Black shale and siltstone

Ke **Etivluk Group (Lower Cretaceous)**—Shale and siltstone

KPs **Sedimentary rocks (Lower Cretaceous to Permian)**—Sandstone and shale

TrPss **Siltstone and shale (Triassic to Permian)**—Sandy siltstone and shale

Trss **Fossiliferous shale (Triassic)**—Black, fossiliferous shale

Volcanic Rocks

Trv **Volcanic rocks, undivided (Triassic)**—Lava flows and lahar deposits that cover large areas of map

Trp **Pyroclastic deposits and rocks**—Basaltic and andesitic pyroclastic rocks and deposits, mostly poorly sorted and nonwelded

Trpt **Coarse-grained ash-flow tuff**—Well sorted, coarse-grained ash-flow tuff. Contains clasts as large as 3 mm

Trpft **Fine-grained to massive ash-flow tuff**—Moderately well sorted, coarse-grained ash-flow tuff. Thickness, 5 m

Trpwt **Welded silicic tuff unit**—Welded silicic tuff made up of fine-grained silicic ash. Thickness, 0.5 m

Trplt **Lithic-rich tuff layer**—Angular lithic clasts. Interbedded with unit Trpft

EXPLANATION OF MAP SYMBOLS

Contact—Solid where location is accurate; dashed where location is approximate; dotted where location is concealed

Gradational contact

Faults—Solid where location is accurate; dashed where location is approximate; dotted where location is concealed

Thrust fault—Sawteeth on upper plate

Detachment faults

Detachment fault within metamorphic core complex block—Hachures on upper plate. Forms boundary beneath unit Trss

Listric fault at head of detachment—Ticks on upper plate

Master detachment fault—Hachures on upper plate

PLANAR POINT FEATURES

[May be combined with other point features at point of observation]

Strike and dip of bedding

40 **Inclined**

Vertical

INTRODUCTION

The rocks and deposits in the map area represent a wide range of stratigraphic, structural, and tectonostratigraphic relations.

Unit ages are from Dalrymple and others (1982).

STRATIGRAPHY

UNCONSOLIDATED DEPOSITS

Surficial deposits in the map area consist of poorly sorted alluvial gravel and lithic sand and silt; also includes large areas of glacial outwash.

BEDROCK

A diverse set of volcanic and sedimentary rocks are present in the map area. These bedrock lithologies represent fragments of several tectonostratigraphic terranes.

KLIPPEN OF ENDICOTT MOUNTAINS AND DE LONG MOUNTAINS SUBTERRANES NEAR MOUNT ANNETTE

Rocks of the Endicott Mountains and DeLong Mountains subterranes are present in several scattered outcrops that are separated by faults.

Endicott Mountains Subterrane

Rocks of the Endicott Mountains subterrane consist of sedimentary and volcanic rocks that range in age from Permian to Lower Cretaceous.

Volcanic Rocks

Volcanic rocks of the Endicott Mountains subterrane are found in the extreme southeastern part of the map area.

Pyroclastic Rocks

Pyroclastic rocks of the Endicott Mountains subterrane primarily consist of welded and nonwelded ash-flow tuffs.

Ash-flow Tuff

A conspicuous layer of laminated, lithic-rich ash-flow tuff is interbedded with unit Trpft.

REFERENCES CITED

Bailey, E.H., Irwin, W.P., and Jones, D.L., 1964, Franciscan and related rocks: California Geological Survey Bulletin 183, 177 p.

Brabb, E.E., Pampeyan, E.H., and Bonilla, M.G., 1972, Landslide susceptibility in San Mateo County, California: U.S. Geological Survey Miscellaneous Field Studies Map MF–360.

Geologic Map of the Palo Alto 7.5′ Quadrangle, California

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